How-To Library

Informatica

Deploying Big Data Management 10.2.2 on the AWS Cloud Platform through the Amazon Marketplace

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Abstract

Customers of Amazon Web Services (AWS) and Informatica can deploy Informatica® Big Data Management 10.2.2 through the AWS marketplace. The automated marketplace solution fully integrates Big Data Management with the AWS platform and the Amazon EMR cluster. The installed solution includes several pre-configured mappings that you can use to discover the capabilities of Big Data Management to load, transform, and write data to various AWS storage resources.

Supported Versions

- Big Data Management 10.2.2

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Overview

Customers of Amazon Web Services and Informatica can deploy a big data solution that fully integrates Big Data Management with the AWS cloud platform and the Amazon EMR cluster.

Several different methods are available for deploying Big Data Management:

Hybrid deployment

Install and configure the Informatica domain and Big Data Management on-premise, and configure them to push processing to the Amazon EMR cluster.

Manual cloud deployment

Manually install and configure the Informatica domain and Big Data Management on AWS EC2 instances in the same region as your Amazon EMR cluster, or deploy the domain on-premises.

Marketplace cloud deployment

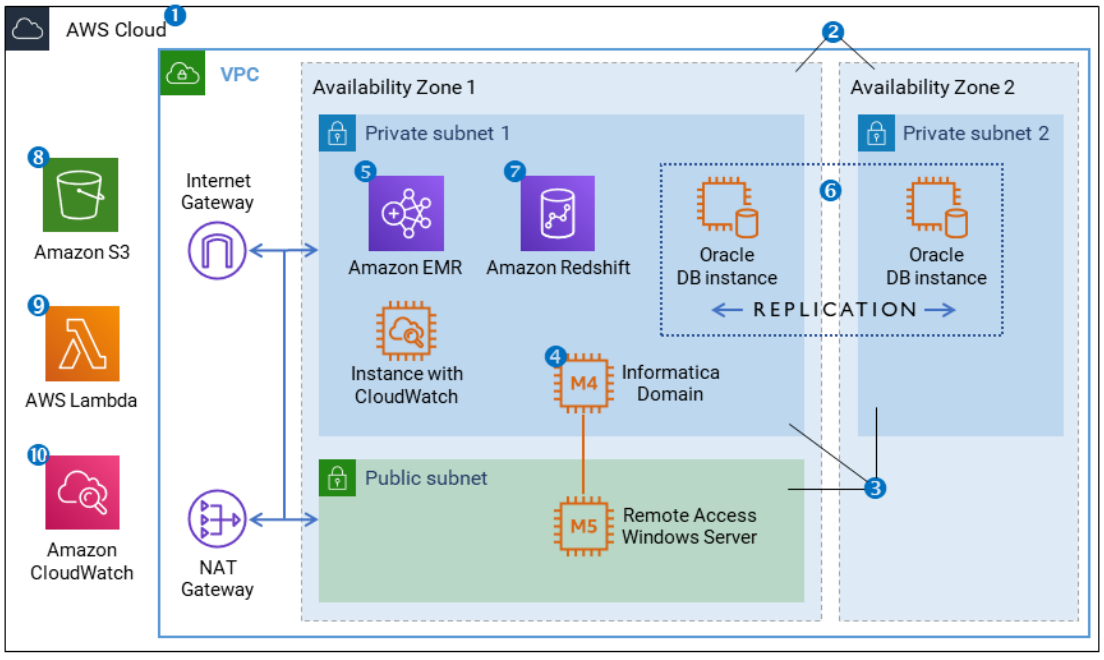
Execute a Big Data Management deployment from the AWS marketplace to create an Informatica domain and an Amazon EMR cluster in the AWS cloud, exploring Big Data Management functionality through prepackaged mappings.

The Big Data Management marketplace solution on AWS creates and connects the following resources in the VPC:

- Informatica domain server on an EC2 instance, with additional instances to contain nodes in the Data Integration Service grid
- Informatica clients on a remote Windows server, on a public subnet
- EMR cluster
- Amazon S3 storage resources, including S3 hosts for source and target data
- Amazon RDS relational databases for Informatica domain repositories and optionally for source and target data
- AWS security and account management services
- AWS regions and Lambda functions

The marketplace solution includes prepackaged mappings that demonstrate various Big Data Management functionality.

The following diagram shows the architecture of the Big Data Management on AWS marketplace solution:



The numbers in the architecture diagram correspond to items in the following list:

1. A virtual public cloud (VPC) to contain the Big Data Management deployment.
2. Availability zones.
3. Subnets to contain specific elements of the deployment. Create two private subnets, plus one public subnet if you want to use a remote Windows server for Informatica clients. Create each of the subnets in a different availability zone.
4. The Informatica domain, including the Model Repository Service and the Data Integration Service.
5. Amazon EMR cluster to process mappings and other jobs from the Data Integration Service.
6. Amazon RDS databases for Informatica domain repositories:
   - Domain repository database
   - Model repository
   - Monitoring Model repository
7. An Amazon Redshift data warehouse, to act as a repository for data sources and targets.
8. S3 storage, to act as a temporary location for files that the data integration service moves between EC2 instances and the EMR cluster.
9. AWS Lambda functions.
10. Amazon CloudWatch.
11. Big Data Management clients in a separate EC2 instance in a public subnet. See "Informatica clients " on page 5 for an explanation of each of these.

The Big Data Management Solution on the AWS Marketplace

The solution includes fully configured AWS resources including S3 storage, Amazon RDS databases for repositories, an Amazon EMR cluster for processing, and an Informatica domain populated with sample data and mappings. Optionally, you can also include an Amazon Redshift data warehouse and a remote Windows server with Informatica clients.

Informatica Domain

The Informatica domain is a server component that hosts application services, such as the Model Repository Service and the Data Integration Service. These services, together with domain clients, enable you to create and run mappings and other objects to extract, transform, and write data.

Application Services

Model Repository Service

The Model Repository Service manages the Model repository. The Model repository stores metadata created by Informatica products in a relational database to enable collaboration among the products. Informatica Developer, the Data Integration Service, and the Administrator tool store metadata in the Model repository.

Data Integration Service

The Data Integration Service is an application service in the Informatica domain that performs data integration tasks for the Developer tool and for external clients.

Metadata Access Service

The Metadata Access Service is an application service that allows the Developer tool to access Hadoop connection information to import and preview metadata. The Metadata Access Service contains information about the Service Principal Name (SPN) and keytab information if the Hadoop cluster uses Kerberos authentication.

The Informatica domain can run several other services. For more information about Informatica services, see the *Informatica Application Service Guide*.

Domain Repositories

Informatica repositories, posted on SQL databases, store metadata about domain objects. Informatica repositories include the following:

Domain configuration repository

The domain configuration repository stores configuration metadata about the Informatica domain. It also stores user privileges and permissions.

Model repository

The Model repository stores metadata for projects and folders and their contents, including all repository objects such as mappings and workflows.

Monitoring Model repository

The monitoring Model repository stores statistics for Data Integration Service jobs. You configure the monitoring Model Repository Service in the domain properties.

In addition to these domain repositories, the solution also requires a repository for Hive metadata. This repository is hosted on an SQL database. It stores Hive table metadata to enable Hadoop operations.

For more information about domain repositories, see the *Informatica Application Service Guide*.

Informatica clients

You can use several different clients with Informatica Big Data Management:

Administrator tool

The Administrator tool enables you to create and administer services, connections, and other domain objects.

Developer tool

The Developer tool enables you to create and run mappings and other objects that enable you to access, transform, and write data to targets.

Command line interface

The command line interface offers hundreds of commands to assist in administering the Informatica domain, creating and running repository objects, administering security features, and maintaining domain repositories.

Remote Windows Server

AWS Windows instance installed with the Developer tool and the command line interface.

Installing the remote Windows server and its clients is optional.

Informatica Connections

You can use the following connections to connect to Amazon S3 storage, the Amazon Redshift data warehouse, and other data repositories:

- AWS_Redshift
- AWS_S3
- BDMSampleConnection
- HADOOP_EMR

- HBASE_EMR
- HDFS_EMR
- HIVE_EMR
- ProfilingWarehouseConnection
- WorkflowConnection

For more information about using these connections, see the "Connections" appendix in the *Big Data Management 10.2.2 User Guide*.

AWS Platform Elements

When you deploy the Big Data Management solution on the AWS cloud platform, the resulting environment consist of many elements. Some of these elements must pre-exist the implementation process, and others are created by the automated implementation.

This section describes all of the possible elements that the final environment can contain.

AWS Building Blocks

Amazon Web Services offers the basic building blocks of storage, networking, and computation, as well as services such as a managed database, big data, and messaging services.

A Big Data Management deployment on EMR can use the following service offerings:

Amazon EC2 instances

Amazon Elastic Compute Cloud (Amazon EC2) instances on Linux/UNIX or Windows hardware residing in Amazon's data centers and accessible through the Amazon Web Services (AWS) cloud.

EC2 instances provide scalable computing capacity in the AWS cloud. You can launch as many or as few virtual servers as you need with zero investment on hardware. You can configure security and networking, and manage storage. Amazon EC2 enables you to scale up or down to handle changes in requirements or spikes in popularity, reducing your need to forecast traffic.

You can deploy Big Data Management on Amazon EC2 with the ability to scale up and scale down the environment based on requirements. Big Data Management can be deployed in a mixed environment that contains on-premises machines and Amazon EC2 instances.

Amazon S3 storage

Amazon Simple Storage Service (S3) is an easy-to-use object storage. You can use it as the repository for databases, including source and target databases and the databases that Big Data Management requires.

Big Data Management provides native, high-volume connectivity to Amazon S3 and support for Hive on S3. It is designed and optimized for big data integration between cloud and on-premise data sources to S3 as object stores. You can use S3 artifacts including datasets, dashboards, and SQL to configure AWS database services and to compute aggregates for datasets.

Amazon Redshift

Amazon Redshift is a cloud-based, fast, fully managed, petabyte-scale data warehouse that makes it simple and cost-effective to analyze all data using existing business intelligence tools. Amazon Redshift is optimized for computationally intensive workloads such as computation of aggregates and complex joins.

Big Data Management can use Amazon Redshift to provide full fact tables, ad-hoc exploration and aggregation, and filtered drill-downs. Informatica’s PowerExchange for Amazon Redshift connector allow users to securely read data from or write data to Amazon Redshift.

Amazon RDS with Oracle Database

Amazon Relational Database Service (Amazon RDS) makes it easy to set up, operate, and scale a relational database in the cloud. It provides cost-efficient and resizable capacity while managing time-consuming database administration tasks, freeing you up to focus on your applications and business. Amazon RDS offers several database engines to choose from; the Big Data Management deployment uses the Oracle database.

Amazon Virtual Private Cloud

Amazon Virtual Private Cloud (Amazon VPC) lets you provision a logically isolated section of the AWS cloud where you can launch AWS resources in a virtual network that you define. You have complete control over your virtual networking environment, including the ability to select your own IP address range, creation of subnets, and configuration of route tables and network gateways.

You can easily customize the network configuration for your Amazon Virtual Private Cloud. For example, you can create a public-facing subnet for your web servers that has access to the Internet, and place your backend systems such as databases or application servers in a private facing subnet with no Internet access. You can leverage multiple layers of security, including security groups and network access control lists, to help control access to Amazon EC2 and EMR instances in each subnet.

AWS Regions and Other Locality Entities

Regions are self-contained geographical locations where AWS services are deployed. Regions have their own deployment of each service. Each service within a region has its own endpoint that you can interact with to use the service.

Regions and Availability Zones

Regions contain availability zones, which are isolated fault domains within a general geographical location. Some regions have more availability zones than others. While provisioning, you can choose specific availability zones or let AWS select them for you.

Lambda Functions

AWS Lambda is a compute platform where you can upload executable code. Lambda stores and executes applications when needed, and scales up or down to meet user needs, with no administrative burden. The applications can contain code in any Lambda-supported language. For more information, see the AWS Lambda documentation.

NAT Gateway

The NAT gateway instance in the public subnet enables instances in the private subnets to connect to the internet or to other AWS services, but prevents the internet from initiating a connection with those instances.

Networking, Connectivity, and Security

Amazon AWS enables the following networking, connectivity, and security features:

Virtual Private Cloud (VPC)

This service lets you provision a logically isolated section of the AWS Cloud where you can launch resources in a virtual network that you define. The VPC provides a network architecture with multiple public and private subnets that span multiple Availability Zones, so that AWS resources can be deployed in highly available configurations.

The VPC has several different configuration options. See the VPC documentation for a detailed explanation of the options and choose based on your networking requirements. You can deploy Big Data Management in either public or private subnets.

Connectivity to the Internet and Other AWS Services

Deploying the instances in a public subnet allows them to have access to the Internet for outgoing traffic as well as to other AWS services, such as S3 and RDS.

Private Data Center Connectivity

You can establish connectivity between your data center and the VPC hosting your Informatica services by using a VPN or Direct Connect. We recommend using Direct Connect so that there is a dedicated link between the two networks with lower latency, higher bandwidth, and enhanced security. You can also connect to EC2 through the Internet via VPN tunnel.

Security Groups

You can define rules for EC2 instances and define allowable traffic, IP addresses, and port ranges. Instances can belong to multiple security groups.

AWS Key Pairs

A key pair is a pair of public and private keys. Amazon EC2 uses public-key cryptography to encrypt and decrypt login information. Public-key cryptography uses a public key to encrypt a piece of data, such as a password, then the recipient uses the private key to decrypt the data.

When you log in to any Amazon EC2 system, you use a password file for authentication. The file is called a private key file and has a file name extension of .pem.

Choose to use an existing key pair or create a new key pair. If you create a new key pair, you save the .pem file to your desktop system, and AWS saves the key pair to your account. Subsequent deployments can reuse the key pair.

For more information about key pairs, see the AWS documentation.

AWS Direct Connect

AWS Direct Connect makes it easy to establish a dedicated network connection from your premises to AWS. Using AWS Direct Connect, you can establish private connectivity between AWS and your datacenter, office, or co-location environment, which in many cases can reduce your network costs, increase bandwidth throughput, and provide a more consistent network experience than Internet-based connections.

Using AWS Direct Connect with the Big Data Management deployment is optional.

Understanding Amazon Cluster Components

The central component of Amazon EMR is the cluster. A cluster is a collection of Amazon Elastic Compute Cloud (Amazon EC2) instances.

Amazon Node Types

Each instance in the cluster is called a node. Each node has a role within the cluster, referred to as the node type. Amazon EMR also installs layers of software components on each node type, giving each node a role in a distributed application like Apache Hadoop.

Amazon EMR nodes are of the following types:

Master node

Manages the cluster by running software components which coordinate the distribution of data and tasks among other nodes—so-called slave nodes—for processing. The master node tracks the status of tasks and monitors the health of the cluster.

Core node

A servant node that has software components which run tasks and store data in the Hadoop Distributed File System (HDFS) on the cluster.

Task node

An optional slave node that has software components which only run tasks.

Amazon EMR Layers

Amazon EMR service architecture consists of several layers, each of which provides certain capabilities and functionality to the cluster. An EMR cluster has the following layers:

Storage

The storage layer includes the different file systems that are used with your cluster. You can choose from among the following storage options:

- Hadoop Distributed File System (HDFS) is a distributed, scalable file system for Hadoop. HDFS distributes the data it stores across instances in the cluster, storing multiple copies of data on different instances to ensure that no data is lost if an individual instance fails. This ephemeral storage that is reclaimed when you terminate a cluster. HDFS is useful for caching intermediate results during MapReduce processing or for workloads which have significant random I/O.
- EMR File System (EMRFS). Amazon EMR uses the EMR File System to enable Hadoop to access data stored in Amazon S3 as if it were a file system like HDFS. You can use either HDFS or Amazon S3 as the file system in your cluster. Most often, Amazon S3 is used to store input and output data, and intermediate results are stored in HDFS.
- Local File System. The local file system refers to a locally connected disk. When you create a Hadoop cluster, each node is created from an Amazon EC2 instance that comes with a pre-configured block of pre-attached disk storage called an instance store. Data on instance store volumes persists only during the life cycle of its Amazon EC2 instance.

Cluster resource management

The resource management layer is responsible for managing cluster resources and scheduling the jobs for processing data.

By default, Amazon EMR uses YARN (Yet Another Resource Negotiator), which is a component introduced in Apache Hadoop 2.0 to centrally manage cluster resources for multiple data-processing frameworks. However, there are other frameworks and applications that are offered in Amazon EMR that do not use YARN as a resource manager. Amazon EMR also has an agent on each node which administers YARN components, keeps the cluster healthy, and communicates with the Amazon EMR service.

Data processing frameworks

The data processing framework layer is the engine used to process and analyze data. Many frameworks run on YARN or have their own resource management.

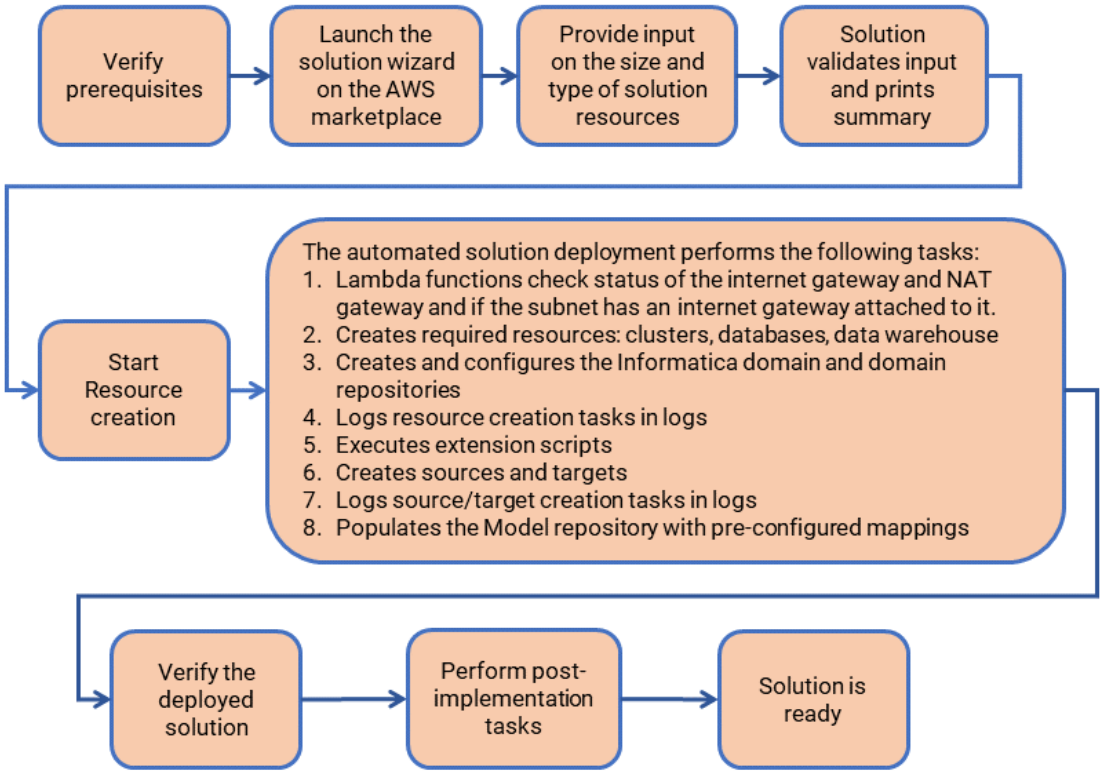
Informatica documentation refers to these data processing frameworks as run-time engines. The engine you choose depends on processing needs, such as batch, interactive, in-memory, or streaming. Your choice of run-time engine affects the languages and interfaces on the application layer, which is the layer used to interact with the data you want to process.

The following main run-time engines are available for Amazon EMR:

- **Spark.** Apache Spark is a cluster framework and programming model for processing big data workloads. Like Hadoop MapReduce, Spark is an open-source, distributed processing system but uses directed acyclic graphs for execution plans and leverages in-memory caching for datasets. Spark supports multiple interactive query modules such as SparkSQL
- **Blaze.** Informatica Blaze is the industry's unique data processing engine integrated with YARN to provide intelligent data pipelining, job partitioning, job recovery, and scalability, which is optimized to deliver high performance, scalable data processing leveraging Informatica's cluster aware data integration technology.

Implementation Overview

The following diagram shows how the marketplace solution is implemented:



Pre-Implementation Tasks

Before you start the automated Big Data Management deployment, perform the following pre-implementation tasks:

- Verify prerequisites.
- Gather information about the AWS account and environment.

Prepare Your AWS Account

To prepare to start the Big Data Management 10.2.2 deployment on AWS, complete the following steps:

1. If you need to create an AWS account for the deployment, create an account at https://aws.amazon.com by following the on-screen instructions.

   Grant the account user full administrative privileges, which are necessary to configure and deploy the solution using AWS services and resources.
2. Use the region selector in the navigation bar to choose the AWS region where you want to deploy Big Data Management.
3. Identify a key pair to use, or create a key pair in the region that you selected. Make a note of the key pair to use for the deployment.

   Make a note of the key pair to use for this deployment.

Gather AWS Account and Environment Information

Before you begin to provision the Big Data Management marketplace solution stack on AWS, gather the following information about the AWS account and environment:

Property	Description
AWS_ACCESS_KEY	AWS access key for the user account to use to initialize the stack.
AWS_SECRET_ACCESS_KEY	AWS secret access key for the user account to initialize the stack.
REGION	AWS region in which to create the stack.
EMR_ROLE	AWS EMR full access role. This role must be assigned to the user account that initializes the stack before this script is executed.
EC2_INSTANCE_PROFILE	AWS ec2 access policy. This policy must be assigned to the user account that initializes the stack.
EC2_KEY_PAIR	2048-bit SSH-2 RSA key pair to use for stack security. For more information about key pair requirements and how to generate the key pair, see Amazon documentation.
EC2_SUBNET	AWS subnet id. This subnet should be present in the VPC in which the stack is created.
MASTER_SECURITY_GROUP	AWS security group for master EMR node. The user account that initializes the stack must be assigned to this security group.
SLAVE_SECURITY_GROUP	AWS security group for slave EMR node. The user account that initializes the stack must be assigned to this security group.
HADOOP_NODE_JDK_HOME	AWS EMR JDK path. Required to modify the Hadoop connection for mappings run in Blaze mode.

Register at Informatica Marketplace

Informatica Marketplace registration and a subscription to Informatica Big Data Management are required to deploy the Big Data Management marketplace solution on AWS.

1. Choose to create a new Informatica marketplace account, or use an existing account.
   - Register at Informatica Marketplace if you are not already registered. Confirm your Informatica Marketplace account through the verification email and input your password when prompted.
   - To use an existing account, log in to Informatica Marketplace using your existing username and password.
2. Subscribe to Big Data Management.

Other Prerequisites

Verify the following prerequisites before you begin to provision the AWS marketplace solution:

- Big Data Management license. The marketplace offering is a "bring your own license" (BYOL) solution. Get the license from the email message from Informatica and upload it to S3 storage on your AWS account.
- AWS roles and privileges. The user should have AWS administrator privileges, or permissions to create IAM roles.
- Verify that VPC peering is enabled on the VPC where the resources that you want to use reside.

Provision the Big Data Management on AWS Marketplace Solution

Use the AWS Marketplace website to provision AWS platform resources including a Big Data Management deployment.

Launch the Implementation Wizard

Log on to the AWS marketplace site and select **Cloud Formation** > **Stacks** > **Create Stack** to begin the process of deploying the solution.

Populate the properties on the **Create Stack** page. The following sections describe the properties to populate.

Specify Details section

In this section, specify a stack name for the solution on the AWS platform.

Network Configuration section

In this section, specify network parameters for the solution.

Several fields ask you to supply CIDR values. The CIDR (Classless Inter-Domain Routing) value represents a block of IP addresses. For example, to specify the range of 10.20.30.0 to 10.20.30.255, enter the following string: `10.20.30.40/24`

Configuration of the private and public subnets depends on whether you want the automated deployment to create a remote Windows server with Informatica clients.

- If you do not want the remote windows server, specify CIDR values for two private subnets and leave the value for the Public Subnet CIDR property blank.
- If you want the remote Windows server, specify CIDR values for two private subnets and one public subnet. For more information about the remote Windows server, see “Informatica clients ” on page 5.

The following table describes values for parameters in the **Network Configuration** section:

Availability Zones	List of Availability Zones to use for the subnets in the VPC. List at least two zones. The solution prioritizes the zones in the order in which you list them.
VPC CIDR	CIDR block for the VPC to use for the deployment. To use existing resources, identify a VPC where existing resources reside. Verify that VPC peering is enabled.
Public Subnet CIDR	CIDR block for the optional public subnet.
Private Subnet 1 CIDR	CIDR block for one of two private subnets.
Private Subnet 2 CIDR	CIDR block for the second of two private subnets.
IP address range	CIDR IP address range that is permitted to access the Informatica domain.
Deploy a Remote Windows Server?	Choose from the following: - No (default). - Yes. The solution deploys a remote Windows server that can access other resources in the VPC. **Note:** If you choose Yes, then choose service subnets which are not attached to an Internet gateway.

Amazon EC2 Configuration section

The following table describes values for parameters in the **Amazon EC2 Configuration** section:

Parameter	Description
Key pair name	Select an existing EC2 KeyPair name to enable SSH access for Informatica services to the EC2 instance.

Amazon RDS Configuration section

In this section, specify parameters for the Amazon RDS relational database to host the Informatica domain. The following table describes values for parameters in this section:

Parameter	Description
Database password	Type a password for the domain repository database user. Retype this password in the next field.
Select MultiAZ deployment	Choose Yes to enable the database to be available in more than one availability zones, or No (default).

Informatica Big Data Management Configuration section

In this section, specify parameters for the Big Data Management domain. The following table describes values for parameters in this section:

Parameter	Description
Informatica administrator user name	Type a name for the Informatica domain administrator. In this field and the following field, you can specify any user name and password. Make a note of the user name and password, and use it later to log in to the Administrator tool to configure the Informatica domain.
Informatica administrator password	Type a password for the Informatica domain administrator. Retype this password in the next field.
Big Data Management License Key Location	Name of the Amazon S3 bucket in your account that contains the Informatica Enterprise Information Catalog Key. Use a bucket of the same region in which the stack is being launched.
Big Data Management License Key Name	Enter the subdirectory path, if any, and filename of the Big Data Management license key file located in the S3 bucket named in the property Big Data Management License Key Location . For example, where the entire path including the bucket name is `S3BucketName/SubDir1/SubDir2/BDMLicense.key`, type the following: `SubDir1/SubDir2/BDMLicense.key`

Amazon EMR configuration

In this section, specify parameters for the Amazon EMR cluster. The following table describes values for parameters in this section:

Parameter	Description
EMR AutoDeploy	Choose Yes to enable autodeploy connections for the solution, or No (default) to create a new EMR cluster and connections.
EMR ID of EMR Cluster	ID of the existing EMR cluster to use in the solution. Choose <NONE> if you want to: - Create a new EMR cluster for the solution - Use cluster workflows (auto-deployment) to create ephemeral clusters for the solution Default: <NONE>

Amazon Redshift configuration

In this section, optionally specify parameters for a cluster for the Amazon Redshift data warehouse.

To create a new Redshift data warehouse, configure the parameters as follows:

- Redshift Deploy Type = Required
- Redshift Host = <NONE>

To use an existing Redshift data warehouse, configure the parameters as follows:

- Redshift Deploy Type = Required
- Redshift Host = <IP address of the existing Redshift cluster master node>

If you do not want to use a Redshift data warehouse with the deployed solution, choose:

- Redshift Deploy Type = Skip

Then click **Next** to go to the next deployment step.

The following table describes values for parameters in this section:

Parameter	Description
Redshift Deploy Type	Choose from the following options: - Required. Choose Required to create a Redshift data warehouse cluster. - Skip. Choose Skip if you do not want to use an Amazon Redshift data warehouse.
Redshift Host	Choose from the following options: - To use an existing Redshift cluster, type the DNS name or IP address of the master node of the existing Redshift cluster. - To create a Redshift cluster, choose <NONE>. Default: <NONE>
Redshift user name	Type the username that is associated with the master user account for the Redshift cluster.
Redshift database name	Type a name for the Redshift data warehouse.
Redshift cluster password	Type a password for the Redshift cluster master user account.

After you finish entering values for the parameters, click **Next**.

AWS begins provisioning resources according to the values you entered.

Monitoring Instance Provision and Informatica Domain Creation

You can use cloud platform dashboards, logs, or other artifacts to see whether cluster creation succeeded and how to locate and identify the Informatica domain on the cloud platform.

AWS Resources

You can monitor the progress of the solution deployment in the Create Stack section of the AWS portal.

The following image shows the display when the solution deployment is complete:

BDMStack2-BDMInformatica-1...	NESTED	2019-03-28 19:30:49 UTC+0550	CREATE_COMPLETE	Informatica Enterprise Infor...
BDMStack2-EMRStack-1RY3K...	NESTED	2019-03-28 19:16:53 UTC+0550	CREATE_COMPLETE	EMR
BDMStack2-RedshiftClusterSta...	NESTED	2019-03-28 19:16:14 UTC+0550	CREATE_COMPLETE	AWS CF Template to create ...
BDMStack2-S3BucketStack-13...	NESTED	2019-03-28 19:16:12 UTC+0550	CREATE_COMPLETE	Informatica S3 Bucket
BDMStack2-RDSStack-EIEJFF...	NESTED	2019-03-28 19:14:21 UTC+0550	CREATE_COMPLETE	This template creates RDS r...
BDMStack2-IAMStack-9N98OF...	NESTED	2019-03-28 19:13:27 UTC+0550	CREATE_COMPLETE	This template creates IAM r...
BDMStack2-SecurityGroupStac...	NESTED	2019-03-28 19:13:26 UTC+0550	CREATE_COMPLETE	Create Informatica EIC Elasti...
BDMStack2-ServiceSubnetChe...	NESTED	2019-03-28 19:12:45 UTC+0550	CREATE_COMPLETE	This template is used to che...
BDMStack2-ServiceSubnetsAZ...	NESTED	2019-03-28 19:12:45 UTC+0550	CREATE_COMPLETE	This template is used to che...
BDMStack2-PublicSubnetChec...	NESTED	2019-03-28 19:12:44 UTC+0550	CREATE_COMPLETE	This template is used to che...

Click the **Output** tab to view the list of resources that the deployment process created in the VPC. The list includes the solution creation log, the location of Informatica administration logs, and the URL to access the Informatica Administrator tool.

The following image shows the list of resources:

Key	Value	Description	Export Name
InstanceSetupLogs	/var/log/cfn-init-cmd.log	Informatica Domain EC2 Instance setup Log...	BDMStack1-BDMStack-1QR6P76HF0RP5-BDMInformatica-1CTOTVK23CMJ9:InstanceSetupLogs
InformaticaAdminConsoleURL	http://ip-18-0-25-168.ec2.internal:6008	Informatica Administrator Console	BDMStack1-BDMStack-1QR6P76HF0RP5-BDMInformatica-1CTOTVK23CMJ9:InformaticaAdminConsoleURL
InformaticaAdminConsoleServerLogs	/home/ec2-user/Infa_OnceClick_Solution.log	Informatica Domain Installation Log Location	BDMStack1-BDMStack-1QR6P76HF0RP5-BDMInformatica-1CTOTVK23CMJ9:InformaticaAdminConsoleServerLogs

Informatica Resources

The list of deployed resources includes the **InformaticaAdminConsoleURL**. Click the linked value of this property.

The following image shows an example of this property:

Key	Value
InformaticaAdminConsoleURL	http://ip-18-0-25-168.ec2.internal:6008

The Administrator tool opens in a browser window.

Use the Informatica Administrator user name and password to log in to the Administrator tool.

You can use the Administrator tool to view and administer the Informatica application services and resources that the automated deployment created.

Solution Deployment Logs

You can review the following logs to see the solution deployment as it happened.

Review the following logs in the `/var/log` directory:

- cfn-init.log
- cfn-init-cmd.log
- cfn-wire.log
- cloud-init.log
- messages

Review the following log in the `/home` directory:

- Infa_OnceClick_Solution.log

Logs

After the completion of Big Data Management deployment, consult logs to see the success or failure of solution element creation.

You can access the following logs on the Informatica domain VM:

Command execution log

This log records the following events:

- Creation of Informatica connections, cluster configurations, and services.
- Population of the data warehouse and SQL databases.
- Import of sample mappings to the Model repository. This is recorded in the Project Importing section of the log.
- Data Integration Service recycling to register all changes.

At the top of the log file is a summary section that lists automated tasks and their status. Beneath the summary section are detailed sections about each task. If any of the tasks failed to complete successfully, you can look at the detailed section for the task to troubleshoot the task.

Filename: `Infa_OnceClick_Solution.log`

Location: `/home`

CloudWatch logs

CloudWatch is an AWS services that enables you to monitor all of the events in your AWS deployments. CloudWatch provides a central location from which to monitor EC2 instances and other resources. for more information, see the AWS documentation.

To monitor Big Data Management events in CloudWatch, go to the CloudWatch page and search for "Informatica."

If no log files are returned by the search, log into the VM where the Informatica domain is installed and check for logs with the filename "`cfn*.log`"

Note: if you configured your Big Data Management implementation in subnets that are not attached to Internet gateways, then the VM will not have a public IP address. To access the logs, create a VM in a subnet that meets the following criteria:

- The subnet is a public subnet.
- The subnet is attached to an internet gateway.

- The subnet is a member of the same VPC as the VPC where you configured Big Data Management.

You can use this public VM to access the domain VM or the administrator console.

Post-Implementation Tasks

After the marketplace solution is finished deploying on AWS, perform the tasks in this section.

Manually Remove the Database Password from the DBcreation.log File

The DBcreation.log file records information about domain databases, including the database user password in clear text. Remove this password for security reasons.

Edit the DBcreation.log file to remove the database user password. The file is located in the following path: /home/ec2-user/individual_logs.

Add the Domain IP Address and DNS Name to the Hosts File

if you did not create a remote Windows server during configuration of the Big Data Management solution, you must add the IP address and DNS name of the Informatica domain to the hosts file on each machine that you use to access the domain.

During configuration of the Big Data Management solution, you have the choice to create a remote Windows server. If you skipped creation of the remote Windows server, you must add an entry for the Informatica domain to the hosts file on each machine that you use to access the domain.

Edit the hosts file to add an entry for the IP address and DNS name of the EC2 instance where the Informatica domain resides:

```
<public IP address of the EC2 instance> <private DNS name of the EC2 instance>
```

For example:

```
123.123.123.123 infa-domain-private-dns.aws.com
```

Using the Pre-Installed Mappings

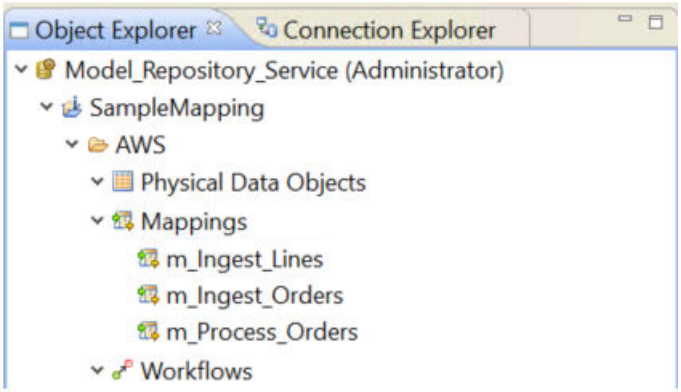
The marketplace solution contains sample pre-configured mappings that you can use as templates for your own mappings.

This section lists and describes the mappings and contains instructions for how to run them.

Pre-Installed Mappings

Use the Developer tool to open and run the pre-installed mappings that the automated deployment contains.

Browse the folders in the BDM_Sample project to access the pre-installed mappings. The following image shows the sample mappings inside the project folder:



The following table lists pre-installed sample mappings that you can run on the Amazon EMR cluster:

Folder	Mapping Name	Demonstrates
Amazon_AWS	m_Ingest_Lines_to_S3	Writing mapping results to S3
Amazon_AWS	m_Ingest_Orders_to_S3	Writing mapping results to S3
Amazon_AWS	m_Process_Orders_S3_to_Redshift	Reading from S3 and writing data to Redshift

Run the Mappings

Browse to the mapping in the Developer tool and double-click the mapping to open it in the editor.

Each of the included mappings is configured to use the Spark engine. You must use the Spark engine to run the pre-configured mappings.

To run a mapping, select it in the list of mappings in the Navigator, and choose **Run** > **Run**.

Author

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